Kiru0310 / Python-for-machine-learning

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Python-for-machine-learning / 240901062-time series-5.ipynb

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188 lines (188 loc) · 4.69 KB

Preview Code Blame Raw

In [5]:
```
import pandas as pd
import numpy as np
from datetime import datetime
dates=[datetime(2020,1,5),
        datetime(2020,1,10),
        datetime(2020,1,15),
        datetime(2020,1,20),
        datetime(2020,1,25)]
ts=pd.Series(np.random.randn(5),index=dates)
print("\nWith this date object,create a series with datetime index:\n",ts)
```

```
With this date object,create a series with datetime index:
 2020-01-05   -0.961048
2020-01-10   -0.425087
2020-01-15    1.771624
2020-01-20    0.468228
2020-01-25    1.039003
dtype: float64
```

In [6]:
```
res=ts.truncate(after="1/15/2020")
print("\nTruncate method to truncate datetime series\n",res)
```

```
Truncate method to truncate datetime series
 2020-01-05   -0.961048
2020-01-10   -0.425087
2020-01-15    1.771624
dtype: float64
```

In [7]:
```
da=pd.date_range("10/01/2022",periods=10,freq="W-SUN")
print("\nDate range with day periods.\n",da)
```

```
Date range with day periods.
 DatetimeIndex(['2022-10-02', '2022-10-09', '2022-10-16', '2022-10-23',
               '2022-10-30', '2022-11-06', '2022-11-13', '2022-11-20',
               '2022-11-27', '2022-12-04'],
              dtype='datetime64[ns]', freq='W-SUN')
```

In [11]:
```
date=pd.DatetimeIndex(["1/1/2020","1/2/2020","1/2/2020","1/3/2020"])
ts1=pd.Series(np.arange(4),index=date)
print("\nDuplicate datetime index:\n",ts1)
```

```
Duplicate datetime index:
 2020-01-01    0
2020-01-02    1
2020-01-02    2
2020-01-03    3
dtype: int64
```

In [12]:
```
group=ts1.groupby(level=0)
print("\nRepeated date counts:\n",group.count())
```

```
Repeated date counts:
 2020-01-01    1
2020-01-02    2
2020-01-03    1
dtype: int64
```

In [13]:
```
long_df=pd.DataFrame(np.random.randn(10,4),index=da,columns=list("ABCD"))
print("\nMulti-column DataFrame with datatime index:\n",long_df)
```

```
Multi-column DataFrame with datatime index:
                   A         B         C         D
2022-10-02  0.611661 -0.826908  0.362463  0.876580
2022-10-09  0.571146 -0.371954 -1.635266 -0.442961
2022-10-16 -0.412978  0.133963 -0.266706 -0.664683
2022-10-23  2.766236 -0.358564 -0.985218  2.800842
2022-10-30 -1.041284  2.325458 -0.116337 -0.055230
2022-11-06  0.047010  0.213705  0.859822 -1.165431
2022-11-13  0.165230 -0.391425  1.932854 -0.445661
2022-11-20 -0.814611  1.333726  1.682034 -0.372378
2022-11-27  1.636071  0.004666 -0.922911 -1.478807
2022-12-04 -0.476485  0.303413  0.359605  0.969159
```